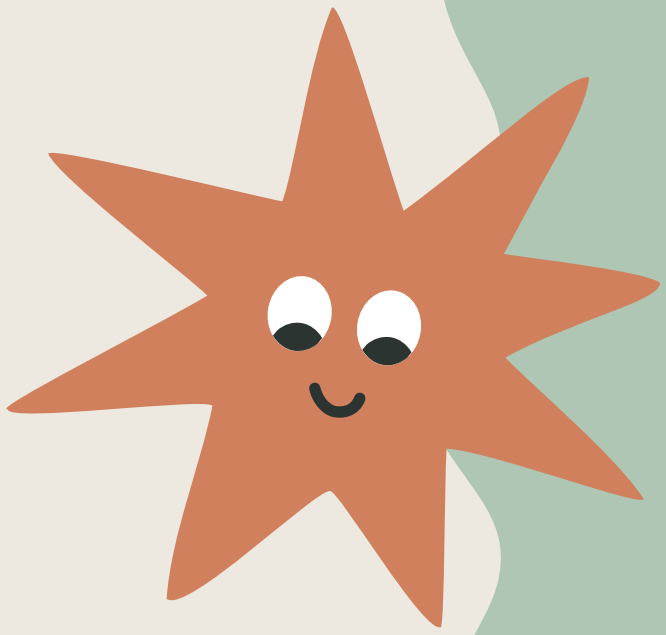
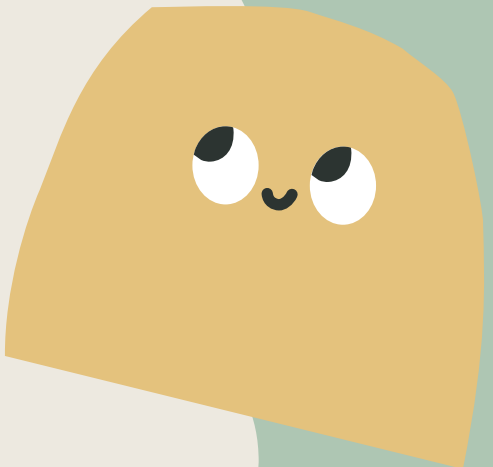
Types of Intelligent Agents and Environments

Lecture 4





Learning Outcomes (Lecture 1)

➢At the end of this lecture, students will be able to:

- Describe the architecture and behavior of a simple reflex agent.
- Explain goal-based agents and goal-oriented decision making.
- Explain utility-based agents and utility functions.
- Describe the components of a learning agent.
- Classify environments according to their characteristics.
- Differentiate between fully and partially observable environments.
- Differentiate between deterministic and stochastic environments.
- Differentiate between static and dynamic environments.
- Differentiate between single-agent and multi-agent environments.
- Implement a simple rule-based agent using Python..

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What is an Agent Architecture?

- Agent architecture describes the basic structure through which an agent receives information, processes it, makes decisions, and produces actions
- For example:
  - Sensors → Processing → Decision → Actuators
- Different agent types use different kinds of processing and decision-making

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Types of Intelligent Agents

- There are five different types of intelligent agents
  1. Simple Reflex Agent
  2. Model-Based Agent
  3. Goal-Based Agent
  4. Utility-Based Agent
  5. Learning Agents
- However, this does **not** mean every real-world AI system fits perfectly into only one category

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1. Simple Reflex Agent

- A simple reflex agent selects an action based only on the current percept
- In simple words:
  - See something → Apply a rule → Perform an action
- It does not normally consider the history of previous percepts

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What is a Rule?

- A rule is simply IF-THEN statement
- For example:

```
IF room is dirty
THEN clean
```

```
IF traffic light = red
THEN stop
```

```
IF temperature > 30°C
THEN turn on air conditioner
```

- Examples include:
  - Automatic doors
  - Simple thermostats
  - Basic alarm systems

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2. Model-Based Agent

- A model agent maintains an internal representation of the world
- Internal state is information the agent remembers about the environment
- For example:

```
Current location = Room B
Room A = Clean
Room B = Dirty
Last direction = Left
```

- Even if the agent cannot currently see Room A, it may remember what it previously observed

Prepared by Nishan Rasheed

3. Goal-Based Agent

- A goal-based agent selects actions based on a desired **goal**
- A **goal** is a desired state or outcome
- Example:
  - Navigation: Reach Islamabad
  - Chess: Checkmate the opponent
  - Delivery robot: Deliver the parcel to Room 205
  - Vacuum Cleaner: Clean all rooms
- The goal tells the agent:
  - Where do I want to end up?

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Goal-Based Decision Making

- Suppose a robot is at Sialkot, goal it to reach Islamabad
- There may be multiple possible actions
- Agent considers which action helps me reach the goal?
- This may require **search**
- Search is an important AI technique where an agent explores possible sequences of actions to find a solution

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4. Utility-Based Agent

- An intelligent agent that selects an action by evaluating the possible outcomes and choosing the action that provides the **highest utility**
- Utility represents how desirable or beneficial an outcome is according to the agent's preferences
- In simple words:
  - Utility tells us how good a particular outcome is
  - We can assign a numeric value
- For example:
  - Outcome A → Utility = 60 and Outcome B → Utility = 90
  - The agent prefers Outcome B

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Destination Example

- Now imagine two possible routes both reach the destination
- Route A
  - 100 km, 2 hours, heavy traffic
- Route B
  - 120 km, 1.5 hours, low traffic
- Which should the agent select?
- A goal alone may not tell us.
- We need to measure **how desirable each outcome is**

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Utility Function

- A **utility function** is a function that assigns a numerical value to an outcome or state according to how desirable it is
- For example:
  - Safe route → 90
  - Fast route → 80
  - Expensive route → 40
  - Dangerous route → 10
- The agent attempts to choose actions that maximize expected utility

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Destination Example Solution

- Step 1: Define what the agent cares about
  - **Time:** 50% importance
  - **Traffic:** 30% importance
  - **Distance:** 20% importance
  - The agent wants **less time, less traffic, and less distance**
- Step 2: Give each route scores
- Let's use a simple scale where **10 = best** and **1 = worst**.

Factor	Route A	Route B
Distance	9	7
Time	6	10
Traffic	3	9

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Destination Example Solution

- Now calculate utility:

$$U(A) = (9 \times 0.20) + (6 \times 0.50) + (3 \times 0.30)$$

$$U(A) = 1.8 + 3.0 + 0.9 = \mathbf{5.7}$$

$$U(B) = (7 \times 0.20) + (10 \times 0.50) + (9 \times 0.30)$$

$$U(B) = 1.4 + 5.0 + 2.7 = \mathbf{9.1}$$

- Step 3: Choose the route with higher utility
- Therefore utility-based agent chooses Route B

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5. Learning Agent

- A learning agent can improve its behavior based on experience
- Instead of relying entirely on rules designed by a programmer, the agent can learn from:
  - Data
  - Experience
  - Feedback
  - Previous outcomes
- Example is creating a system to detect cat images

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Comparing the Five Agent Types

Agent Type	Main Idea	Memory	Goal	Learning
Simple Reflex	Current percept → rule → action	No	No	No
Model-Based	Percept + internal state	Yes	Not necessarily	No
Goal-Based	Choose actions to achieve goal	Usually	Yes	No
Utility-Based	Choose best/most desirable outcome	Usually	Yes	No
Learning Agent	Improve using experience	Yes	May use goals/utility	Yes

Characteristics of Environment

- There are six different characteristics of environment
  1. Fully observable vs partially observable
  2. Deterministic vs stochastic
  3. Episodic vs sequential
  4. Static vs dynamic
  5. Discrete vs continuous
  6. Single-agent vs multi-agent

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1. Fully Observable vs Partially Observable

➢Full Observable Environment

- An environment is full observable when the agent's sensors provide all information necessary to determine the relevant current state of the environment
- The agent can see everything important that it needs to make its decision
- Example: Chess

➢Partially Observable Environment

- An environment is partially observable when the agent cannot access all relevant information about the current state
- Example: Self-driving car

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2. Deterministic vs Stochastic

- An environment is deterministic if the next state is completely determined by the current state and the agent's action
- In simple words:
  - Same situation + same action → same result
  - Example: Chess
- An environment is stochastic if there is uncertainty about what will happen after an action
  - Example: Self-driving car changing lane

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3. Episodic vs Sequential

- An environment is episodic when each decision/action is relatively independent of previous decisions
  - Example: Image classification
- A sequential environment is one where the current action can affect future situations
  - Example: Chess, Driving

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4. Static vs Dynamic

- A static environment does not change while the agent is deciding what to do
  - Example: A crossword puzzle
- A dynamic environment can change while the agent is making a decision
  - Example: Self-driving car

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5. Discrete vs Continuous

- An environment is discrete when states, actions or time steps can be represented as separate, countable values
  - Example: Chess
  - The board consists of a finite arrangement of pieces
- A continuous environment has values that can change smoothly over a range
  - Example: Self-driving car
  - Speed can be: 40 km/h
  - 40.1 km/h
  - 40.2 km/h
  - 40.3 km/h

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6. Single-Agent vs Multi-Agent

- A single-agent environment contains one primary agent making decisions
  - Example: A puzzle-solving AI
- A multi-agent environment contains multiple agents whose actions can affect one another
  - Examples: Chess, Football-playing robots, Online games

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Environment Classification Examples

Environment	Observable	Deterministic	Dynamic	Agents
Chess	Fully	Mostly deterministic	Mostly static	Multi-agent
Crossword puzzle	Fully	Deterministic	Static	Single-agent
Self-driving	Partially	Stochastic	Dynamic	Multi-agent
Image classification	Usually fully for the input	Usually deterministic	Static	Single-agent
Robot navigation	Often partially	Stochastic	Dynamic	May be multi-agent
Online multiplayer game	Often partially	Often stochastic	Dynamic	Multi-agent

Why Environment Matters

- Suppose Agent A is working in:
  - Fully observable, deterministic, static environment.
  - It can use relatively simple decision-making
- Suppose Agent B is working in:
  - Partially observable, stochastic, dynamic environment
  - It needs much more sophisticated techniques
  - It may require
    - Internal state, prediction, probability, planning, learning
- The complexity of the environment strongly influences the design of the intelligent agent

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Exercise

- Identify agent types in the following examples:
- Scenario 1:
  - IF temperature > 30
  - THEN turn AC ON
- Answer: Simple Reflex Agent
- Scenario 2:
  - A robot remembers that a room was previously dirty even though it cannot currently see the room
- Answer: Model-Based Agent

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Exercise

- An autonomous taxi receives information from cameras, GPS, radar, and traffic sensors. It maintains an internal map of nearby vehicles and roads. Its objective is to reach the passenger's destination safely. When several routes are available, it chooses the route that balances travel time, fuel consumption, safety, and toll cost. The taxi also learns from previous trips to improve future route selection.
- **Identify:**
  - Agent type
  - Observable or partially observable?
  - Deterministic or stochastic?
  - Episodic or sequential?
  - Static or dynamic?
  - Discrete or continuous?
  - Single-agent or multi-agent?

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Solution

Property	Answer	Reason
Agent type	**Learning Agent**	It learns from previous trips
Observability	**Partially Observable**	Sensors cannot see everything
Determinism	**Stochastic**	Traffic and other drivers create uncertainty
Episodes	**Sequential**	Current decisions affect future decisions
Environment	**Dynamic**	Traffic and vehicles continuously change
State	**Continuous**	Speed, position, distance, etc.
Agents	**Multi-Agent**	Other drivers/pedestrians interact

Important: Although this taxi also uses a utility function, the best classification is **Learning Agent** because learning is a defining capability

Exercise

- A medical AI receives a patient's symptoms, medical history, laboratory results, and imaging data. Some information about the patient's condition is unavailable. The system considers several possible diagnoses and recommends the diagnosis that has the highest probability. It continuously updates its beliefs when new test results become available.
- **Identify the environment**

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Solution

- **Agent type: Utility-based / knowledge-based reasoning agent**, depending on the implementation
- **Partially observable** → complete patient condition cannot be directly observed.
- **Stochastic** → the same symptoms can correspond to different diseases.
- **Sequential** → new tests and previous results affect future decisions.
- **Dynamic** → patient's condition can change over time.
- **Single-agent** → the AI is the primary decision-making agent.

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Exercise

- A smart thermostat monitors room temperature and turns heating or cooling on/off. It only uses the current temperature reading and follows these rules:

```
IF temperature < 20°C
    Turn Heater ON

IF temperature > 22°C
    Turn Heater OFF
```

- It doesn't remember previous temperatures, doesn't plan ahead, and doesn't learn

Prepared by Mamun Rasheed

Solution

- Agent type → **Simple Reflex Agent**
- Observable/Partially Observable → **Mostly Fully Observable**
- Deterministic/ Stochastic → **Mostly Deterministic**
- Episodic/Sequential → **Episodic**
- Static/Dynamic → **Dynamic** if room temperature can change while deciding
- Discrete/Continuous → **Continuous**
- Single/Multi-agent → **Single-Agent**

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Exercise

- An AI trading agent observes market prices, trading volume, news, and other indicators. It cannot know exactly what will happen next. It can buy, sell, or hold stocks. Its objective is to maximize long-term return while controlling risk
- Identify agent type and environment?

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Solution

- Agent type → **Utility-Based**
- Observable → **Partially Observable**
- Deterministic → **Stochastic**
- Sequential → **Sequential**
- Static → **Dynamic**
- Discrete/Continuous → **Continuous**
- Single/Multi-agent → **Multi-Agent**

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Solution Explanation

➢Why Multi-Agent?

- The market contains many other participants whose actions influence prices

➢Why Utility-Based?

- The agent is not simply trying to achieve:
  - “Buy a Stock”
- It evaluates outcomes using factors such as:
  - Return + Risk + Cost
- and tries to maximize overall utility

Prepared By: Ehtisham.Rasheed

Exercise

- An autonomous delivery robot operates inside a warehouse.
  - It has cameras and distance sensors.
  - It maintains an internal map.
  - It knows its delivery destination.
  - It chooses the shortest safe path.
  - Workers can move around unexpectedly.
  - The robot improves its navigation after repeated deliveries.
- Identify agent type and environment?

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Solution

Property	Answer
Agent type	**Learning Agent**
Observability/Partially Observable	**Partially Observable**
Determinism/Stochastic	**Stochastic**
Episodic/Sequential	**Sequential**
Static/Dynamic	**Dynamic**
Discrete/Continuous	**Continuous**
Single/Multi-Agent	**Multi-Agent**